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Page count:	21
Word count:	10,369
Character count:	56,897
Submission date:	15-Jul-2026 11:26PM (UTC-0400)
Submission ID:	2998683868

A Depth-Aware, Physics-Based Data Augmentation Framework for Simulating Rain and Fog Distortions in Outdoor RGB Images

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Abstract

Outdoor RGB driving datasets often underrepresent adverse-weather conditions such as rain and fog, making it difficult to train distortion-aware detection models and evaluate robustness systematically. This paper presents a depth-aware, physics-based augmentation framework for simulating rain and fog in outdoor driving images. Using BDD100K as the baseline dataset, the framework generates rainy images through image-plane rendering, depth-aware physics-based rendering using depth maps produced by Depth Anything V2, Cycle-GAN translation, and a hybrid Cycle-GAN plus physics-based pipeline. For fog simulation, a Cycle-GAN generator trained on Cityscapes and Foggy Cityscapes is applied to BDD100K while preserving object-detection labels. The augmented datasets are evaluated by training YOLOv8n detectors and testing them on clear, rainy, and foggy BDD100K validation splits. Results show that replacing clean training data with augmented data generally reduces clear-weather and rainy-weather detection performance, whereas combining clean and augmented data preserves performance close to the clean-data baseline. The strongest gains are observed on the foggy split, although this finding remains preliminary due to its limited foggy sample size. Overall, the framework shows that controlled adverse-weather augmentation is most effective as a supplement to clean training data and provides a practical foundation for future robustness studies.

1 Introduction

Images used in computer vision are often distorted. Distortion in computer vision refers to any signal that produces a visual deviation in an image/video from its expected appearance (Gillani et al., 2025). When the original image/video is unavailable, Gillani et al. (2025) also consider "observed visual comfort or simply annoying effect on the aesthetic visual appearance" to be a distortion. These distortions can be sorted into In-Capture Distortion, which occurs during the initial acquisition phase, including physical artifacts like blur, noise, uneven illumination, color shifts, haze, rain, snow, and other visibility changes; and Post-Capture Distortion, which refers to image deformation or degradation that occurs after images are captured, such as transmission impairments, enhancement artifacts, and adversarial distortion.

Although these effects may appear minor to human observers, they can change the statistical structure of an image and reduce the reliability of machine learning models trained primarily on clean data (Gillani et al., 2025). Distortions affect the performance of high-level tasks, such as object classification, object detection, and scene understanding (Gillani et al., 2025). High-level tasks are more sensitive to distortions than low-level tasks, such as edge detection, image enhancement, and image colorization, because models often learn specific, localized semantic features rather than global context. Therefore, distortion detection and removal are possible preprocessing steps for high-level tasks to decrease the impact on performance. Figure 1 demonstrates several common distortions in high-level tasks.

For machine learning, distortion is not only an image-quality issue but is also affected by the dataset and training. Models usually learn from the distribution of inputs they are given, so a model trained only on clean images may have poor generalizability when processing the same object appearing under poor visibility, weather effects, or degraded contrast. Distortion-specific datasets are important because they make it possible to evaluate model robustness by distortion type and severity, rather than reporting only a single clean-data accuracy score (Gillani et al., 2025).

Current datasets of natural distortions have important limitations. Real rain, haze, fog, and snow images can be difficult to collect at scale, and the resulting data may be unbalanced across weather type, severity, scene category, and annotation quality. Many benchmark datasets also include distortions as broad corruptions rather than as physically meaningful, scene-dependent effects. This limits how well they can support controlled robustness studies on deep